

Tail-Spans and Differential Transcendence

in Surreal and Surcomplex
Hahn Fields

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Abstract

We give an exact algebraic classification of finite tuples of Hahn series whose coefficients use independent square roots. The decisive invariant is not the full span of the coefficient vectors but their *cofinite span*: the subspace that remains after every finite deletion. If its dimension is d and E is the finite set of exceptional vectors, the generated algebra is exactly a polynomial algebra in d variables over the multiquadratic extension belonging to E . Consequently, both the transcendence degree and all geometric components of the relation ideal are explicit.

A finite-difference argument converts independent coefficient sign changes into vanishing conditions on the highest homogeneous part of a hypothetical polynomial relation. This yields a continuum-sized, explicitly indexed family of differentially algebraically independent *actual surreal numbers*, all with real-algebraic coefficients and positive integer valuation support. In particular, the Berarducci–Mantova differential transcendence degree of $\mathbb{A}_{\mathbb{R}}((t^{\mathbb{Q}}))$ over $\mathbb{Q}((t^{\mathbb{Q}}))$ is $2^{\aleph_0}$, with $t = \omega^{-1}$.

For surcomplex analysis, the same mechanism proves that every mixed jet of $\sum_{n \in A} t^n \sqrt{1 - z/2^n}$ is algebraically independent over $\mathbb{C}(z)((t^{\mathbb{Q}}))$, for every infinite $A \subseteq \mathbb{N}_{>0}$, with respect to d/dz and $t \, d/dt$. A continuum-sized family enjoys joint mixed-jet independence on one common disk. We also give exact algebraic-approximation criteria, finite witnesses, and limitations on coefficient-stream decision procedures.

Research status. Complete mathematical proofs are supplied relative to standard Hahn-field and surreal normal-form foundations. The tail-span classification and the explicit differential-independence constructions are proposed contributions; a targeted literature search did not identify these formulations, but priority is not certified. No named historical conjecture is claimed solved. The article is not independently refereed or proof-assistant verified. Accompanying exact computations test finite identities and examples, not the infinite theorems.

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1 The research question and the contributions

An infinite Hahn sum can be far more algebraically complicated than any of its coefficients. This is especially relevant to surreal mathematics: a Conway normal form may use only real-algebraic coefficients and very simple exponents, while the resulting number is not controlled by any finite algebraic or differential equation over a substantial Hahn subfield. For analytic coefficient functions the discrepancy is stronger still: every coefficient may satisfy a quadratic equation on the same disk, while the whole function and all of its mixed derivatives satisfy no polynomial relation over a large rational-coefficient Hahn field.

The aim here is an exact structural result, not merely the production of one transcendental series. We ask how all polynomial relations in a finite family of such series can be read from the coefficient vectors. The answer is a finite-dimensional invariant with a finite algebraic exceptional part. This produces explicit differential independence, including inside the actual surreal differential field.

1.1 The main statements in compact form

Let M be a field of characteristic zero, let Γ be an ordered abelian group, and let $a_i \in M^\times$ have independent square classes. Choose $r_i^2 = a_i$ and distinct exponents γ_i whose set is well ordered. For $v_i \in M^r$, put

$$Y = \sum_{i \in I} t^{\gamma_i} r_i v_i, \quad B = M((t^\Gamma)), \quad W = \bigcap_{F \subseteq I \text{ finite}} \text{span}_M \{v_i : i \notin F\}. \quad (1.1)$$

Theorem 5.1 proves that $E = \{i : v_i \notin W\}$ is finite and

$$B[Y_1, \dots, Y_r] = M(r_i : i \in E)((t^\Gamma))[s_1, \dots, s_d], \quad d = \dim_M W, \quad (1.2)$$

where the s_j are algebraically independent over the displayed coefficient field. The equality is an equality of subrings in a specified ambient Hahn field, not merely an abstract comparison of dimensions. Theorem 5.2 then identifies the full relation ideal after its finite splitting extension.

For actual surreal numbers, write

$$B_0 = \mathbb{Q}((t^\mathbb{Q})), \quad H_0 = \mathbb{A}_\mathbb{R}((t^\mathbb{Q})), \quad t = \omega^{-1},$$

where $\mathbb{A}_\mathbb{R}$ is the field of real-algebraic numbers. We construct a family $(\xi_\alpha)_{\alpha \in \{0,1\}^\mathbb{N}} \subset H_0$ such that

$$\{D_{\text{BM}}^k \xi_\alpha : \alpha \in \{0,1\}^\mathbb{N}, k \geq 0\} \text{ is algebraically independent over } B_0. \quad (1.3)$$

Thus $\text{dtrdeg}_{B_0} H_0 = \mathfrak{c}$, where $\mathfrak{c} = 2^{\aleph_0}$ (Theorem 7.3). This is a differential result about specified set-sized subfields of $\mathbf{No}$, not an appeal to the size of the whole proper class.

The analytic counterpart starts with the single explicit function

$$F(z, t) = \sum_{n \geq 1} t^n \sqrt{1 - z/2^n}. \quad (1.4)$$

Every square root is the branch equal to 1 at $z = 0$. On one common disk, all mixed jets $\partial_z^j \delta^k F$, with $\delta = t d/dt$, are algebraically independent over $\mathbb{C}(z)((t^\mathbb{Q}))$ (Theorem 9.3). Theorem 10.1 supplies continuum many jointly independent such functions.

1.2 What was checked in the repository

The repository was read at revision

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The main README and documentation catalogue were inspected, together with the opening spectral-theory article and the Laurent-birthday report inventory [1, 2]. The catalogue distinguishes common-domain analytic Hahn functions from coefficientwise germs and separates coefficientwise summation from the fine topology of $\mathbf{No}[i]$. Those conventions are retained here.

The inspected catalogue already organizes substantial material on polynomial algebra, analytic geometry, residues, differential equations, rank-one geometry, and finite spectral theory. The present article instead centers on *algebraic relations among infinitely many algebraic coefficients*, and on the differential transcendence they generate. This is a targeted coverage audit, not a claim to have read every archived manuscript or every Lean declaration. In particular, an unfinished Lean bridge is not treated as an open mathematical theorem.

1.3 Attribution and novelty boundaries

Hahn support calculus, finite separable base change, and multiquadratic Galois theory are background, not claimed innovations. We give proofs of the algebraic ingredients to make the new argument independently auditable; standard Galois and Kummer references are the Stacks Project [5, 6]. Generalized-series operator methods have an extensive literature, including van der Hoeven [7]. Conway normal forms and their use as Hahn fields are reviewed, for example, by Kaplan–Krapp–Serra [8]. The existence, strong additivity, real constant field, and normalization $D_{\mathbf{BM}}\omega = 1$ of the derivation used here are due to Berarducci–Mantova [9].

The proposed contribution is the *combined exact classification* in (1.2), its elementary polarization proof, and the explicit ordinary and mixed differential-independence constructions. Differential transcendence itself is a classical subject; methods based on differential fields and functional equations, for example, appear in Mijajlović–Malešević [10]. No claim is made that the general idea of producing transcendence from growing coefficient extensions is new. The searches were targeted rather than exhaustive, and neither a priority theorem nor a resolution of a previously named open problem is asserted.

1.4 Two places where this report meets its companions

The restriction of the Berarducci–Mantova derivation to a rational-exponent Hahn workspace is *not* proved here. The companion report on differential algebra and differential equations [3] fixes that workspace and proves that formula, and Section 7 cites it. Only the further restriction from real to real-algebraic coefficients, which is immediate, is checked here.

In the other direction, the companion report on dynamics and normal forms [4] carries a warning, `dyn:warn:diffalg`, that its own transcendental solutions must *not* be called differentially transcendental, since they satisfy a first-order linear equation over its base $\mathbb{C}(w)((t^\Gamma))$. Theorem 9.3 obtains the differential conclusion that warning declines, for different functions, over the $\Gamma = \mathbb{Q}$ case of a base of the same shape and on one fixed disk with no shrinking. Remark 9.4 states the relation precisely and does not weaken the warning, which remains true of the functions it is about.

2 Hahn fields, coefficient extensions, and summation

2.1 Set-sized conventions

All fields, groups, index families, and vector spaces in the algebraic proofs are sets. A Hahn series over a field M with value group Γ is a formal expression

$$f = \sum_{\gamma \in S} c_\gamma t^\gamma, \quad c_\gamma \in M, \quad S \subseteq \Gamma \text{ well ordered.}$$

Its support consists of the exponents with nonzero coefficient. Addition is coefficientwise, multiplication is convolution, and $v(f) = \min \text{supp}(f)$ for $f \neq 0$, with $v(0) = +\infty$. The resulting field is denoted $M((t^\Gamma))$.

A family is *Hahn summable* when its supports have well-ordered union and only finitely many family members contribute at each exponent. This is a support condition. It does not mean convergence of a sequence of partial sums in the fine topology of the entire surreal field. We use the standard Hahn support facts: finite sums of well-ordered subsets are well ordered, each fixed exponent has only finitely many representations from a pair of such supports, and the monoid generated by a well-ordered positive set has the corresponding finite-word property. These are the usual Neumann support lemmas; their operator uses are treated in [7].

For $\Gamma = \mathbb{Q}$, the map

$$\sum_q c_q t^q \mapsto \sum_q c_q \omega^{-q} \quad (2.1)$$

embeds $\mathbb{R}((t^\mathbb{Q}))$ into $\mathbf{No}$ and, coefficientwise in real and imaginary parts, $\mathbb{C}((t^\mathbb{Q}))$ into $\mathbf{No}[i]$. This is an instance of the standard Conway normal-form correspondence [8]. No construction of the entire surreal field, or of its real closedness, is part of this paper.

2.2 Finite coefficient extensions commute with Hahn series

Lemma 2.1 (Finite base change). *Let F/M be a finite field extension of degree e , with basis $u_1, \dots, u_e$. Inside a common coefficient extension,*

$$F((t^\Gamma)) = M((t^\Gamma))F, \quad [F((t^\Gamma)) : M((t^\Gamma))] = e.$$

The same $u_1, \dots, u_e$ form a basis after Hahn base change. If F/M is finite Galois, its automorphisms act coefficientwise and give the full Galois group of the base-changed extension.

Proof. Expand each coefficient of $f \in F((t^\Gamma))$ in the M -basis: $c_\gamma = \sum_j m_{j,\gamma} u_j$. For each j , the support of $\sum_\gamma m_{j,\gamma} t^\gamma$ is contained in $\text{supp}(f)$. Thus $f = \sum_j u_j f_j$ with $f_j \in M((t^\Gamma))$. Conversely, a finite sum of this kind is an F -coefficient Hahn series. If $\sum_j u_j f_j = 0$, coefficient comparison and the original basis independence imply that every f_j is zero. This proves the degree and basis assertions. Each coefficient automorphism preserves supports and convolution, and the e distinct coefficient automorphisms exhaust the Galois group when F/M is Galois. $\square$

Warning 2.2. For an infinite algebraic extension L/M , the analogous equality $L((t^\Gamma)) = M((t^\Gamma))L$ generally fails. The field on the right is the union of the finite constant extensions from Lemma 2.1; a Hahn series on the left can involve coefficients generating an infinite extension. Section 3 makes the obstruction exact, and Section 7 shows that the failure can have maximal differential transcendence degree in a countable-coefficient workspace.

2.3 Independent square classes and independent sign changes

Definition 2.3. A family $(a_i)_{i \in I} \subset M^\times$ has *independent square classes* if, for every nonempty finite $J \subseteq I$, $\prod_{i \in J} a_i$ is not a square in M . Fix square roots r_i in an algebraic closure and put $L = M(r_i : i \in I)$.

Lemma 2.4 (Independent sign changes). *For finite $J \subseteq I$, the extension $M(r_i : i \in J)/M$ has degree $2^{|J|}$ and basis $\{\prod_{i \in U} r_i : U \subseteq J\}$. Every assignment $r_i \mapsto \varepsilon_i r_i$, $\varepsilon_i \in \{1, -1\}$, extends to an M -automorphism of L . It extends coefficientwise to an automorphism of $L((t^\Gamma))$ fixing $M((t^\Gamma))$.*

Proof. We give the usual elementary multiquadratic argument. Suppose the claim holds for $r_1, \dots, r_m$, and write $E_m = M(r_1, \dots, r_m)$. If $u^2 = a_{m+1}$ for some $u \in E_m$, each sign automorphism σ satisfies $\sigma(u) = \pm u$. Hence u is a simultaneous eigenvector for the sign group. The basis monomials r_U have distinct sign characters, so their simultaneous eigenspaces are exactly the one-dimensional spaces Mr_U . Thus $u = cr_U$, contradicting independence of the square classes. Adjoining r_{m+1} therefore doubles the degree. The old sign changes extend by fixing r_{m+1} , and the new sign change fixes E_m . This proves the finite assertion by induction, starting with $m = 0$.

Every element of L belongs to a finite subextension. The automorphisms given by one fixed sign assignment are compatible on all such subextensions and define an automorphism of their union. Coefficientwise extension to Hahn series preserves zero coefficients, support, addition, and the finite convolution sums, proving the last assertion. $\square$

Remark 2.5 (No order-preserving automorphism is asserted). When $M = \mathbb{Q}$ and the r_i are positive real square roots, a sign change need not preserve the order of the multiquadratic subfield. We use it only as a field automorphism of that subfield and its Hahn field. We do *not* claim that it extends to an automorphism of the ordered field of all real-algebraic numbers, or to an automorphism of **No**. The proofs below need neither extension.

3 The exact coefficient-field criterion

The following preliminary is a useful classical Galois-orbit argument. It isolates the scalar algebraicity issue before the genuinely multivariable classification.

Theorem 3.1 (Coefficient-field criterion). *Let M have characteristic zero, let $\overline{M}$ be an algebraic closure, and let*

$$f = \sum_{\gamma} c_{\gamma} t^{\gamma} \in \overline{M}((t^{\Gamma})), \quad F = M(c_{\gamma} : \gamma \in \text{supp}(f)), \quad B = M((t^{\Gamma})).$$

Then f is algebraic over B if and only if F/M is finite. In that case

$$B(f) = F((t^{\Gamma})), \quad [B(f) : B] = [F : M]. \quad (3.1)$$

Consequently, the relative algebraic closure of B inside $\overline{M}((t^{\Gamma}))$ is

$$\mathcal{A} = \bigcup_{F/M \text{ finite}, F \subseteq \overline{M}} F((t^{\Gamma})). \quad (3.2)$$

Proof. Assume first that F/M is finite. Lemma 2.1 gives $[B(f) : B] \leq [F : M]$. Every M -embedding of F into $\overline{M}$ extends to an automorphism of $\overline{M}$; this is the standard extension property for algebraic closures in separable characteristic zero [5]. Distinct embeddings differ on a coefficient of f , since those coefficients generate F . Their coefficientwise extensions therefore give $[F : M]$ distinct conjugates of f over B . The reverse degree inequality follows. Since $B(f) \subseteq F((t^{\Gamma}))$ and both extensions have the same finite degree, equality holds.

Conversely, suppose f has degree D over B . Choose any finite subextension F_0/M generated by finitely many coefficients of f . Its distinct embeddings into $\overline{M}$ extend as above and produce at least $[F_0 : M]$ distinct conjugates of f . Hence $[F_0 : M] \leq D$. The degrees of all such F_0 form a bounded nonempty set of positive integers. Choose one, F_* , of maximal degree. Adjoining any coefficient c_{γ} gives another finite coefficient-generated field and cannot increase that degree. Therefore $c_{\gamma} \in F_*$ for every γ , so $F = F_*$ is finite. Formula (3.2) follows immediately. $\square$

Example 3.2. Let p_n be the n th prime and choose the positive roots. Then

$$x = \sum_{n \geq 1} \sqrt{p_n} t^n$$

is transcendental over $\mathbb{Q}((t^{\mathbb{Q}}))$: its coefficients generate an infinite multiquadratic extension. Every finite truncation is algebraic, and its first N terms generate a degree- 2^N constant extension after Hahn base change. Section 7 proves the substantially stronger assertion that x is differentially transcendental and belongs to large explicit families of differentially independent surreals.

4 Cofinite spans and the polarization mechanism

4.1 The span that survives every finite deletion

Let $(v_i)_{i \in I}$ be any family in a finite-dimensional M -space V . No finite alphabet, boundedness, periodicity, or computability hypothesis is imposed on the v_i .

Definition 4.1 (Cofinite span). The *cofinite span* of the family is

$$W = \bigcap_{F \subseteq I \text{ finite}} \text{span}_M \{v_i : i \notin F\}.$$

The term “tail-span” will also be used. Here “tail” means cofinite: for an arbitrary ordered support it does not mean deletion of an arbitrary transfinite initial segment.

Lemma 4.2 (Stabilization and exceptions). *There is a finite $F_0 \subseteq I$ such that $W = \text{span}_M \{v_i : i \notin F_0\}$. The set*

$$E = \{i \in I : v_i \notin W\}$$

is finite. Moreover, every cofinite subfamily of $(v_i)_{i \in I \setminus E}$ spans W .

Proof. Among the dimensions of spans obtained by finite deletion, choose a minimum, attained by F_0 . Deleting any additional finite set from that family gives a subspace of the same dimension, hence the same subspace. For any finite F , the span obtained by deleting $F \cup F_0$ is therefore the F_0 -span and is contained in the F -span. This proves that the intersection equals the F_0 -span. In particular every $i \notin F_0$ has $v_i \in W$, so $E \subseteq F_0$. Finally, deleting E and any other finite set leaves a family contained in W whose span contains the defining intersection W . Its span is exactly W . $\square$

4.2 Multilinear forms detected by distinct indices

Lemma 4.3 (Cofinite spanning detects multilinear forms). *Suppose every cofinite subfamily of $(w_i)_{i \in J}$ spans a finite-dimensional space W . Let K/M be a field extension, and let T be a D -linear form on $W \otimes_M K$, with $D \geq 1$. If*

$$T(w_{i_1}, \dots, w_{i_D}) = 0 \quad \text{whenever } i_1, \dots, i_D \text{ are pairwise distinct,}$$

then $T = 0$.

Proof. If $W = 0$, there is nothing to prove. Otherwise J is infinite, since a finite family would leave the zero span after a finite deletion. Fix distinct $i_1, \dots, i_{D-1}$. The remaining vectors span W , and therefore span its scalar extension over K . Linearity frees the last argument: the expression vanishes for any value of that argument. Now fix only $D - 2$ distinct indices and an arbitrary last argument. The possible next-to-last indexed vectors outside these indices still span W , so that argument is also free. Repeating this procedure frees all D arguments. For $D = 1$ the first spanning step already finishes the proof. $\square$

Lemma 4.4 (Top-degree mixed difference). *Let $P \in K[X_1, \dots, X_d]$ have total degree D , with homogeneous part P_D . For vectors $u_1, \dots, u_D \in K^d$, define $\Delta_u P(X) = P(X + u) - P(X)$. Then*

$$\Delta_{u_1} \cdots \Delta_{u_D} P(X) = D_{u_1} \cdots D_{u_D} P_D = D! T(u_1, \dots, u_D), \quad (4.1)$$

where D_u is directional differentiation and T is the symmetric polarization satisfying $P_D(X) = T(X, \dots, X)$. The result is independent of X .

Proof. Taking one difference lowers total degree by at least one. Its highest remaining homogeneous part is the directional derivative of the previous highest part. After D differences, terms of degree below D have vanished, and only the D successive directional derivatives of P_D remain. This proves the first equality. Expanding $P_D(\lambda_1 u_1 + \cdots + \lambda_D u_D)$ and extracting the coefficient of $\lambda_1 \cdots \lambda_D$ gives the second. Characteristic zero allows division by $D!$, and identifies a homogeneous polynomial uniquely with its symmetric polarization. $\square$

4.3 Why coefficient sign changes defeat polynomial relations

Proposition 4.5 (Sign-orbit independence). *Let $B = M((t^\Gamma))$, let $(a_i)_{i \in I}$ have independent square classes, and let $F \subset I$ be finite. Set $B_F = M(r_i : i \in F)((t^\Gamma))$. Suppose $w_i \in M^d$ for $i \notin F$ and every cofinite subfamily of these vectors spans M^d . If the γ_i are distinct with well-ordered image, then the coordinates of*

$$s = \sum_{i \notin F} t^{\gamma_i} r_i w_i$$

are algebraically independent over B_F .

Proof. For $d = 0$ the assertion is the empty independence assertion. Otherwise, suppose $P(s) = 0$ for a nonzero polynomial $P \in B_F[X_1, \dots, X_d]$ of positive degree D . A nonzero constant cannot vanish, so all possible relations have this form. Put $b_i = t^{\gamma_i} r_i$.

For each $i \notin F$, the coefficient sign change σ_i fixes B_F and sends s to $s - 2b_i w_i$. Distinct such sign changes commute and leave one another's b_i unchanged. Applying their products to $P(s) = 0$ shows that

$$P\left(s - 2 \sum_{j \in J} b_{i_j} w_{i_j}\right) = 0 \quad \text{for every } J \subseteq \{1, \dots, D\},$$

whenever $i_1, \dots, i_D$ are distinct and outside F . Take the alternating sum of these 2^D identities. By Lemma 4.4, it is

$$0 = (-2)^D D! \left(\prod_{j=1}^D b_{i_j} \right) T(w_{i_1}, \dots, w_{i_D}), \quad (4.2)$$

where T is the polarization of P_D , over the coefficient field B_F . All the scalar factors preceding T are nonzero. Thus T vanishes on all distinct indexed tuples. Lemma 4.3 gives $T = 0$, contradicting $P_D \neq 0$. No infinite sequence of Galois operations, and no limiting argument, is used: a relation of degree D is destroyed by finitely many sign changes. $\square$

5 The tail-span classification

5.1 The generated algebra and its algebraic part

Theorem 5.1 (Exact tail-span classification). *Let M have characteristic zero and let Γ be a set-sized ordered abelian group. Let $(a_i)_{i \in I} \subset M^\times$ have independent square classes, choose $r_i^2 = a_i$, and put*

$L = M(r_i : i \in I)$. Let $\gamma_i \in \Gamma$ be distinct with well-ordered image and let $v_i \in M^r$. In $H = L((t^\Gamma))$, define

$$Y = \sum_i t^{\gamma_i} r_i v_i, \quad B = M((t^\Gamma)), \quad W = \bigcap_{F \subseteq I \text{ finite}} \text{span}_M \{v_i : i \notin F\}.$$

Set $d = \dim_M W$, $E = \{i : v_i \notin W\}$, and

$$p = \sum_{i \in E} t^{\gamma_i} r_i v_i, \quad B_E = M(r_i : i \in E)((t^\Gamma)).$$

Then E is finite. For any M -basis of W , let $s_1, \dots, s_d$ be the coordinates of $Y - p$. These coordinates satisfy

$$B[Y_1, \dots, Y_r] = B_E[s_1, \dots, s_d], \quad (5.1)$$

and are algebraically independent over B_E . In particular,

$$B(Y_1, \dots, Y_r) = B_E(s_1, \dots, s_d), \quad (5.2)$$

$$\text{trdeg}_B B(Y_1, \dots, Y_r) = d, \quad (5.3)$$

$$[B_E : B] = 2^{|E|}. \quad (5.4)$$

The relative algebraic closure of B in $B(Y_1, \dots, Y_r)$ is exactly B_E .

Proof. By Lemma 4.2, E is finite and the vectors outside E lie in W , with every cofinite subfamily spanning W . In coordinates on W , Proposition 4.5 proves algebraic independence of $s_1, \dots, s_d$ over B_E . It remains to show that all of B_E is already present in $B[Y_1, \dots, Y_r]$; this is stronger than the transcendence-degree assertion alone.

If E is empty, $B_E = B$ and the assertion is immediate. Otherwise pass to the quotient M^r/W . Each v_i with $i \in E$ has nonzero image. There is an M -linear functional $\ell : M^r \rightarrow M$ vanishing on W and nonzero on every such v_i . Indeed, a finite union of proper hyperplanes cannot exhaust the dual of a finite-dimensional vector space over the infinite field M . Thus

$$\eta = \ell(Y) = \sum_{i \in E} \ell(v_i) t^{\gamma_i} r_i \in B[Y_1, \dots, Y_r] \cap B_E. \quad (5.5)$$

The extension B_E/B is multiquadratic of degree $2^{|E|}$ by Lemmas 2.1 and 2.4. Every sign assignment on E gives a different conjugate of η : at an index where two assignments differ, the coefficient of its distinct exponent γ_i is nonzero and changes sign. Hence η has $2^{|E|}$ distinct conjugates, and $B(\eta) = B_E$.

Because η is algebraic over the field B , its polynomial algebra $B[\eta]$ is already the field $B(\eta)$. Consequently $B_E \subseteq B[Y_1, \dots, Y_r]$. All coordinates of p therefore belong to this generated algebra, and the s_j , being M -linear coordinates of $Y - p$, do as well. This proves the inclusion from right to left in (5.1). The reverse inclusion follows from $Y = p + Us$, where the columns of the matrix U are the chosen basis vectors of W . The field and degree assertions follow.

Finally, a purely transcendental extension $K(s_1, \dots, s_d)$ contains no elements algebraic over K except K itself. For completeness, if a nonconstant rational function depends on some s_j , regard it as a nonconstant rational function in that variable over the field of the others; the variable is then algebraic over that rational function, which proves the rational function is transcendental over that field. An element algebraic over B is also algebraic over B_E , so it must belong to B_E . Conversely every element of B_E is algebraic over B . $\square$

5.2 All geometric components of the relation ideal

Theorem 5.2 (Relation ideal as disjoint affine planes). *In Theorem 5.1, let*

$$I_Y = \ker(B[Z_1, \dots, Z_r] \longrightarrow H, Z_j \mapsto Y_j).$$

Choose an M -linear surjection $q : M^r \rightarrow M^{r-d}$ with kernel W . For $\varepsilon \in \{1, -1\}^E$, put

$$p_\varepsilon = \sum_{i \in E} \varepsilon_i t^{\gamma_i} r_i v_i.$$

Then

$$I_Y B_E[Z_1, \dots, Z_r] = \bigcap_{\varepsilon \in \{1, -1\}^E} (q(Z) - q(p_\varepsilon)). \quad (5.6)$$

Each ideal on the right is generated by $r - d$ independent affine linear forms. The $2^{|E|}$ ideals are pairwise comaximal. Thus the base-changed coordinate algebra is the product of $2^{|E|}$ polynomial algebras in d variables over B_E .

Proof. Theorem 5.1 identifies the coordinate algebra over B with $B_E[s_1, \dots, s_d]$. Finite Galois base change gives

$$B_E \otimes_B B_E \cong \prod_{\sigma \in \text{Gal}(B_E/B)} B_E, \quad a \otimes b \mapsto (a\sigma(b))_\sigma.$$

Adjoining the independent s_j and using $Y = p + Us$, the component indexed by σ sends Z to $\sigma(p) + Us$. Its kernel is precisely $(q(Z) - q(\sigma(p)))$. The sign assignments index all σ , proving (5.6).

If $\varepsilon \neq \varepsilon'$, then $q(p_\varepsilon) - q(p_{\varepsilon'}) \neq 0$. To see this, choose an index where the signs differ. At its distinct exponent the coefficient vector is a nonzero multiple of $q(v_i)$, and $q(v_i) \neq 0$ by the definition of E . At least one quotient coordinate therefore gives two distinct constants. Subtracting the corresponding affine generators gives a nonzero element of the coefficient field B_E , hence a unit. The ideals are comaximal. The Chinese remainder theorem gives the stated product; in particular the components are disjoint and reduced. $\square$

Remark 5.3. Over B , the ideal I_Y is prime, because it is the kernel of a map to a field. Its decomposition after B_E -base change is not a contradiction: the finite algebraic constant field splits into its conjugate factors. The cofinite span determines the affine dimension, while the exceptional vectors determine the entire finite splitting degree.

6 Examples, exact relations, and finite witnesses

6.1 One exceptional coefficient produces a quadratic relation

Take $I = \mathbb{N}_{>0}$, $\gamma_n = n$, $v_1 = (1, 0)$, and $v_n = (1, 1)$ for $n \geq 2$. Put $b_n = t^n r_n$ and $S = \sum_{n \geq 2} b_n$. Then

$$Y_1 = b_1 + S, \quad Y_2 = S, \quad W = M(1, 1), \quad E = \{1\}.$$

The classification becomes

$$B[Y_1, Y_2] = B(r_1)[S], \quad \text{trdeg}_B B(Y_1, Y_2) = 1.$$

The complete relation ideal is

$$I_Y = ((Z_1 - Z_2)^2 - t^2 a_1). \quad (6.1)$$

Indeed, using $U = Z_1 - Z_2$ and $V = Z_2$, the quotient is $B[U]/(U^2 - t^2 a_1)[V]$, and the quadratic is irreducible because r_1 remains quadratic after finite base change. Over $B(r_1)$ it splits into the two parallel lines $Z_1 - Z_2 = \pm t r_1$.

This example explains why simply declaring the tuple “independent up to finitely many terms” would lose important information. The finite term is precisely what creates the nontrivial algebraic constant field.

6.2 Indicator families and recurrence patterns

For subsets $A_1, \dots, A_r \subseteq \mathbb{N}_{>0}$, let

$$Y_j = \sum_{n \in A_j} t^n r_n, \quad v_n = (\mathbf{1}_{A_1}(n), \dots, \mathbf{1}_{A_r}(n)).$$

Only finitely many vector patterns are possible. The cofinite span is exactly the span over M of the patterns occurring infinitely often. Patterns occurring only finitely often can all be removed, whereas an infinitely recurrent pattern survives every finite deletion. Thus

$$\text{trdeg}_B B(Y_1, \dots, Y_r) = \dim_M \text{span}\{\text{infinitely recurrent patterns}\}.$$

Equivalently, this is the rank of the indicator sequences modulo the subspace of finitely supported sequences. For example, pairwise almost disjoint infinite sets give full rank, because each coordinate unit vector occurs infinitely often outside the finite overlaps.

6.3 Vandermonde vectors need no recurring finite alphabet

For $v_n = (1, n, \dots, n^{r-1})$ and distinct positive integers n , every r such vectors are independent. Their determinant is the nonzero Vandermonde product $\prod_{u < v} (n_v - n_u)$. Therefore every infinite subfamily has cofinite span M^r . Theorem 5.1 gives

$$\sum_n t^n r_n, \quad \sum_n n t^n r_n, \quad \dots, \quad \sum_n n^{r-1} t^n r_n$$

as an algebraically independent tuple over $M((t^\Gamma))$ whenever the positive integer exponents are available and the chosen index set is infinite. These are exactly the successive Euler derivatives used below.

6.4 Finite witnesses against a bounded-degree relation

Corollary 6.1. *In Proposition 4.5, suppose $d > 0$. For any degree $D \geq 1$, choose D pairwise disjoint finite index sets, each supplying a basis of M^d . Then a nonzero polynomial of degree D cannot vanish at every point in the sign orbit of s generated by these indices. Some choice of one index from each basis yields a nonzero D -fold mixed difference. Thus at most dD sign-change indices suffice for such a witness.*

Proof. If the highest polarization vanished on every choice of one basis vector from each of the D bases, multilinearity would make it vanish on all of $(M^d)^D$. Choose a tuple where it does not vanish. The indices are pairwise distinct because the bases have disjoint index sets. Equation (4.1) then gives a nonzero alternating sum of 2^D orbit evaluations. Such disjoint bases exist because every cofinite subfamily spans M^d . $\square$

This is a degree-sensitive finite certificate inside the proof. It is not an algorithm deciding arbitrary infinite coefficient streams: locating tail-spanning bases or certifying that they persist forever is additional information. Section 13 makes that distinction explicit.

7 Actual surreal numbers of maximal differential independence

7.1 A specified differential Hahn workspace

Let $\mathbb{A}_\mathbb{R} = \overline{\mathbb{Q}} \cap \mathbb{R}$ be the countable field of real-algebraic numbers. Via (2.1), regard

$$B_0 = \mathbb{Q}((t^\mathbb{Q})) \subset H_0 = \mathbb{A}_\mathbb{R}((t^\mathbb{Q})) \subset \mathbf{No}, \quad t = \omega^{-1}, \quad (7.1)$$

as concrete subfields of the surreal field. The Euler derivation is

$$\delta \left(\sum_q c_q t^q \right) = \sum_q q c_q t^q. \quad (7.2)$$

It preserves supports up to deletion, so it is well defined and strongly additive. The identity $q_1 + q_2 = q$ in each convolution coefficient proves the Leibniz rule.

The normalized Berarducci–Mantova derivation D_{BM} on $\mathbf{No}$ has real constant field, is strongly additive, and satisfies $D_{\text{BM}}\omega = 1$ [9]. Its restriction to a rational-monomial Hahn workspace is not reproved here. The companion report on differential algebra and differential equations [3] already fixes that workspace: its section “The rational-exponent workspace” introduces $\mathbb{C}((t^{\mathbb{Q}}))$ and $\mathbb{R}((t^{\mathbb{Q}}))$ with $t = \omega^{-1}$ at its equation `diff:eq:HahnQ`, and its Proposition `diff:prop:hahnderiv`, at its equation `diff:eq:hahnderiv`, proves for every additive subgroup $\Gamma \subseteq \mathbb{R}$ with $1 \in \Gamma$ that $\mathbb{C}((t^{\Gamma}))$ is D_{BM} -stable, has constant field exactly $\mathbb{C}$, and satisfies

$$D_{\text{BM}} \left(\sum_q c_q t^q \right) = - \sum_q q c_q t^{q+1}.$$

Taking $\Gamma = \mathbb{Q}$ and comparing with (7.2) gives

$$D_{\text{BM}}(t^q) = -qt^{q+1}, \quad D_{\text{BM}}|_{H_0} = -t\delta. \quad (7.3)$$

That proposition is stated for the complex coefficient field, which contains ours; the further restriction to H_0 and B_0 needs only that $-t\delta$ multiplies each coefficient by a rational number and shifts its exponent by 1, so $\mathbb{A}_{\mathbb{R}}$ coefficients stay in $\mathbb{A}_{\mathbb{R}}$ and $\mathbb{Q}$ coefficients stay in $\mathbb{Q}$. Thus both B_0 and H_0 are differential subfields. Their constant fields are respectively $\mathbb{Q}$ and $\mathbb{A}_{\mathbb{R}}$: coefficient comparison in $\delta f = 0$ leaves only exponent zero. The large differential transcendence degree below is therefore not caused by adjoining transcendental constants. This restriction uses only rational powers; no unproved formula for the derivative of an arbitrary Conway monomial is needed. The same companion report’s warning `diff:warn:monomialmap` records why the Conway monomial map $\gamma \mapsto \omega^{-\gamma}$ must not be silently replaced by $\exp(-\gamma \log \omega)$ at a general surreal exponent γ ; both readings agree at the ordinary rational exponents used here, and nowhere below is a monomial with a non-real exponent differentiated.

Induction, or evaluation on every monomial followed by strong additivity, gives the useful triangular identity

$$D_{\text{BM}}^k = (-1)^k t^k \delta(\delta + 1) \cdots (\delta + k - 1), \quad k \geq 0. \quad (7.4)$$

At $k = 0$, the product is the identity. The leading coefficient in terms of δ^k is the nonzero base-field element $(-1)^k t^k$. For every finite collection of elements and every finite jet order, their D_{BM} -jets and Euler jets are related by invertible triangular linear transformations over B_0 .

7.2 One number and every infinite subsequence

Let p_n be the n th prime and $r_n = \sqrt{p_n} > 0$. Unique factorization in $\mathbb{Z}$ shows that the square classes of the p_n are independent in $\mathbb{Q}^{\times}/\mathbb{Q}^{\times 2}$: a nonempty finite product has an odd exponent at every selected prime. For $A \subseteq \mathbb{N}_{>0}$, define the actual surreal number

$$\xi_A = \sum_{n \in A} \sqrt{p_n} \omega^{-n} = \sum_{n \in A} r_n t^n. \quad (7.5)$$

It is positive and infinitesimal when A is nonempty.

Theorem 7.1 (Every infinite subseries is differentially transcendental). *For every infinite $A \subseteq \mathbb{N}_{>0}$, the family $\{\delta^k \xi_A : k \geq 0\}$ is algebraically independent over B_0 . The same is true of $\{D_{\text{BM}}^k \xi_A : k \geq 0\}$.*

Proof. For any $K \geq 0$,

$$(\xi_A, \delta \xi_A, \dots, \delta^K \xi_A) = \sum_{n \in A} t^n r_n(1, n, \dots, n^K).$$

Every $K + 1$ distinct coefficient vectors are a Vandermonde basis over $\mathbb{Q}$. Consequently every cofinite subfamily spans $\mathbb{Q}^{K+1}$. Apply Theorem 5.1 with $W = \mathbb{Q}^{K+1}$ and $E = \emptyset$. It gives independence of the finite jet tuple. Every polynomial in the infinite jet family uses only finitely many jets, proving independence of the whole family. Formula (7.4) transfers this independence to the D_{BM} -jets. $\square$

Remark 7.2. The proof took place in the intermediate multiquadratic Hahn field $\mathbb{Q}(\sqrt{p_n} : n \geq 1)((t^{\mathbb{Q}}))$. Its coefficient sign automorphisms need not extend to H_0 or $\mathbf{No}$; Remark 2.5 applies. Polynomial independence proved in this intermediate field remains true for the same elements in H_0 and $\mathbf{No}$, over the same base B_0 .

7.3 An explicit continuum-sized almost disjoint family

For a nonempty binary string $s = (s_1, \dots, s_m)$, define

$$N(s) = 2^m + \sum_{j=1}^m s_j 2^{m-j}. \quad (7.6)$$

Strings of length m use the integers from 2^m through $2^{m+1} - 1$, so this is an injective coding of all nonempty finite binary strings. For an infinite binary sequence $\alpha = (\alpha_1, \alpha_2, \dots)$, set

$$A_\alpha = \{N(\alpha_1, \dots, \alpha_m) : m \geq 1\}. \quad (7.7)$$

Each A_α is infinite. Two different sequences share only the codes of their common initial prefixes, hence have finite intersection. In a finite collection of distinct sequences, each corresponding set therefore has infinitely many indices belonging to no other set in the collection. We call these its private indices.

Theorem 7.3 (Explicit maximal differential transcendence). *Put $\xi_\alpha = \xi_{A_\alpha}$ using (7.5) and (7.7). Then*

$$\{D_{\text{BM}}^k \xi_\alpha : \alpha \in \{0, 1\}^{\mathbb{N}}, k \geq 0\} \quad (7.8)$$

is algebraically independent over $B_0 = \mathbb{Q}((t^{\mathbb{Q}}))$. In particular, with the induced Berarducci–Mantova derivation,

$$\text{dtrdeg}_{B_0} H_0 = \text{trdeg}_{B_0} H_0 = \mathfrak{c}. \quad (7.9)$$

The map sending free ordinary differential indeterminates X_α to ξ_α embeds the differential polynomial algebra $B_0\{X_\alpha : \alpha \in \{0, 1\}^{\mathbb{N}}\}$ into H_0 .

Proof. It suffices first to prove independence of the Euler jets belonging to finitely many distinct sequences $\alpha_1, \dots, \alpha_r$, up to order K . Their combined coefficient vector at n is

$$v_n = (\mathbf{1}_{A_{\alpha_a}}(n) n^k)_{1 \leq a \leq r, 0 \leq k \leq K} \in \mathbb{Q}^{r(K+1)}.$$

On the infinitely many private indices of A_{α_a} , all blocks except block a are zero. Within that block the vectors are Vandermonde vectors. Every cofinite subfamily of these private indices spans the whole a th block. Doing this for all a shows that the cofinite span of the full combined family is $\mathbb{Q}^{r(K+1)}$. Theorem 5.1 therefore proves independence of the combined finite jet tuple. The triangular change (7.4) gives the same result for D_{BM} -jets. Since any polynomial involves only finitely many sequences and jet orders, (7.8) follows.

There are $\mathfrak{c}$ binary sequences and the ξ_α are pairwise distinct, so this gives differential transcendence degree at least $\mathfrak{c}$. The matching upper bound is a separate cardinality count that uses none of the independence just proved: $\mathbb{A}_\mathbb{R}$ and $\mathbb{Q}$ are countable, and a Hahn series in H_0 is in particular a function from $\mathbb{Q}$ to the countable set $\mathbb{A}_\mathbb{R}$, so $|H_0| \leq \aleph_0^{\aleph_0} = \mathfrak{c}$. A transcendence degree is bounded by the cardinality of the field, so both displayed degrees are at most $\mathfrak{c}$; the exhibited family supplies the reverse inequality for the differential one, and a differentially independent family is in particular algebraically independent. This also gives $|H_0| = \mathfrak{c}$. Finally, a differential polynomial is an ordinary polynomial in finitely many formal jets. Its evaluation is injective precisely by the proved jet independence. $\square$

Remark 7.4 (The index family is not arbitrary). Theorem 7.3 is about the binary-prefix family (A_α) of (7.7), not about all subsets of $\mathbb{N}_{>0}$. The unrestricted family $(\xi_A)_{A \subseteq \mathbb{N}_{>0}}$ is *not* independent: for any A and B , $\mathbf{1}_{A \cup B} + \mathbf{1}_{A \cap B} = \mathbf{1}_A + \mathbf{1}_B$, so

$$\xi_{A \cup B} + \xi_{A \cap B} = \xi_A + \xi_B,$$

already a $\mathbb{Q}$ -linear relation. Almost disjointness is what removes such relations, by giving each member infinitely many private indices; the same role is played in the proof by the cofinite span. Theorem 7.1 is the complementary statement that no hypothesis at all beyond infinitude is needed when only *one* A is considered.

7.4 Why the base field and the derivation matter

The theorem is not merely independence over $\mathbb{Q}(\omega)$. The base B_0 contains *every* rational-coefficient Hahn series with rational valuation support, including supports with infinitely many exponents below a finite rational bound. Nevertheless the displayed numbers escape all finite algebraic differential equations over that entire field. Conversely, independence over B_0 does not mean independence over the whole H_0 , which already contains them.

Let $\mathcal{A}_0$ be the relative algebraic closure of B_0 in H_0 . By Theorem 3.1, it is the union of $F((t^\mathbb{Q}))$ over real number fields $F \subseteq \mathbb{A}_\mathbb{R}$. It is stable under D_{BM} . The same jet family is independent even over $\mathcal{A}_0$. Indeed, adjoining an algebraic base extension does not change the transcendence degree of a finite tuple: the compositum is algebraic over the field of that tuple. Thus

$$\text{dtrdeg}_{\mathcal{A}_0} H_0 = \mathfrak{c}.$$

In particular, every individual coefficient $\sqrt{p_n}$ is already in the algebraic base, but the infinite sum is not differentially algebraic there. Algebraic closure and unrestricted Hahn formation are decisively different operations.

8 Analytic coefficient fields and two commuting derivations

8.1 A common disk and independent quadratic functions

Let $M = \mathbb{C}(z)$ and, for $n \geq 1$, define

$$a_n(z) = 1 - \frac{z}{2^n}, \quad h_n(z) = \sqrt{1 - z/2^n}, \quad h_n(0) = 1. \quad (8.1)$$

Each branch is holomorphic on the disk $D_2 = \{|z| < 2\}$; the first possible branch point is at 2. We usually work on the smaller common disk $D = \{|z| < 1\}$, which also accommodates the dilation identity below. The field

$$L = \mathbb{C}(z)(h_n : n \geq 1)$$

is realized inside the field $\mathcal{M}_0$ of meromorphic germs at 0. The common ambient Hahn field $\mathcal{M}_0((t^{\mathbb{Q}}))$ contains both $L((t^{\mathbb{Q}}))$ and the ring $\mathcal{O}(D)((t^{\mathbb{Q}}))$; neither latter object is asserted to contain the other. This choice of germs fixes compatible square-root choices; finite rational combinations are meromorphic germs even when they do not have a common global domain.

Lemma 8.1 (Independent analytic square classes). *The a_n have independent square classes in $\mathbb{C}(z)^{\times}/\mathbb{C}(z)^{\times 2}$.*

Proof. For any nonempty finite J , the product $\prod_{n \in J} (1 - z/2^n)$ has a simple zero at each 2^n with $n \in J$. A square rational function has even order at every point, since zeros and poles of its square have twice the original integer orders. The product is therefore not a square. $\square$

Our ambient and base fields are

$$H = L((t^{\mathbb{Q}})), \quad B = \mathbb{C}(z)((t^{\mathbb{Q}})), \quad K_{\mathbb{Q}} = \mathbb{C}((t^{\mathbb{Q}})). \quad (8.2)$$

The smaller rational function field $K_{\mathbb{Q}}(z)$ embeds in B . Indeed, the map from $K_{\mathbb{Q}}[z]$ is injective by comparing Hahn coefficients: a polynomial relation in z would make every ordinary complex polynomial coefficient vanish. It therefore extends to the fraction field.

Warning 8.2 (A large relation field is not a common-domain ring). An arbitrary element of $B = \mathbb{C}(z)((t^{\mathbb{Q}}))$ can have coefficient poles approaching 0. Hence B is not being identified with a ring of functions on one common disk. Algebraic relations over B are interpreted in the formal Hahn field H with meromorphic-germ coefficients. The specific functions constructed here *do* lie in $\mathcal{O}(D)((t^{\mathbb{Q}}))$. A passage to actual common-domain or halo functions is made only where justified in Section 11. This distinction agrees with the ring distinctions in the repository catalogue [2].

8.2 The derivations and their exact coefficient formula

The ordinary derivation $\partial = \partial_z$ of $\mathbb{C}(z)$ extends to L : by implicit differentiation of $h_n^2 = a_n$,

$$\partial h_n = \frac{h_n}{2(z - 2^n)} \in L.$$

Define on H

$$\partial \left(\sum_q f_q t^q \right) = \sum_q f'_q t^q, \quad \delta \left(\sum_q f_q t^q \right) = \sum_q q f_q t^q. \quad (8.3)$$

Both maps preserve supports up to deletion. Coefficientwise finite convolution proves the Leibniz rule, and coefficient comparison proves that they commute. They preserve B as well. The same formulas also define commuting derivations on $\mathcal{M}_0((t^{\mathbb{Q}}))$ and preserve its common-domain holomorphic coefficient subring $\mathcal{O}(D)((t^{\mathbb{Q}}))$.

Let

$$c_j = (1/2)_j = \frac{1}{2} \left(\frac{1}{2} - 1 \right) \cdots \left(\frac{1}{2} - j + 1 \right), \quad c_0 = 1.$$

Every c_j is nonzero. Repeated differentiation gives

$$\partial^j h_n(z) = c_j (z - 2^n)^{-j} h_n(z). \quad (8.4)$$

For example, differentiating the right side at order j multiplies it by $(1/2 - j)/(z - 2^n)$, giving the next falling factorial.

For every subset $A \subseteq \mathbb{N}_{>0}$, set

$$F_A(z, t) = \sum_{n \in A} t^n h_n(z) \in H \cap \mathcal{O}(D)((t^{\mathbb{Q}})). \quad (8.5)$$

The support is a subset of the positive integers, so this is a Hahn series without any coefficient growth condition. Its mixed jets have the particularly useful form

$$\partial^j \delta^k F_A = \sum_{n \in A} t^n h_n(z) c_j \frac{n^k}{(z - 2^n)^j}. \quad (8.6)$$

9 Every mixed jet is algebraically independent

9.1 An elementary exponential-graph lemma

Lemma 9.1. *If $R(X, Y) \in \mathbb{C}[X, Y]$ is nonzero, then $R(n, 2^n) = 0$ for only finitely many positive integers n .*

Proof. Write $R(X, Y) = \sum_{m=0}^M p_m(X) Y^m$, with $p_M \neq 0$. If $M = 0$, this is the fact that a nonzero polynomial has finitely many roots. In general, for sufficiently large real n , $p_M(n) \neq 0$ and

$$\frac{R(n, 2^n)}{p_M(n) 2^{Mn}} = 1 + \sum_{m < M} \frac{p_m(n)}{p_M(n)} 2^{-(M-m)n}.$$

Each rational function of n in the sum has at most polynomial growth, while its exponential factor tends to zero. The displayed ratio tends to 1, so it is nonzero for all sufficiently large n . Only finitely many positive integers remain. $\square$

The lemma is an asymptotic statement about an ordinary finite polynomial; it invokes no transcendence conjecture and no assertion about the arithmetic nature of individual values 2^n .

9.2 Every infinite subfamily spans every finite jet space

Lemma 9.2 (Hereditary full jet span). *Fix $J, K \geq 0$. For $n \geq 1$, form the vector*

$$v_n(z) = \left(c_j \frac{n^k}{(z - 2^n)^j} \right)_{0 \leq j \leq J, 0 \leq k \leq K} \in M^{(J+1)(K+1)}, \quad M = \mathbb{C}(z).$$

Every infinite subfamily of these vectors spans the entire displayed space over M . In particular, every cofinite subfamily of every infinite subfamily has full span.

Proof. Suppose an infinite index set A failed to span. A nonzero M -linear functional would annihilate all its vectors. Absorb the nonzero c_j into its coefficients and clear their common rational denominator. There would exist polynomials $C_{jk}(z) \in \mathbb{C}[z]$, not all zero, such that

$$\sum_{j,k} C_{jk}(z) \frac{n^k}{(z - 2^n)^j} = 0 \quad \text{in } \mathbb{C}(z), \quad n \in A. \quad (9.1)$$

Choose the largest j_* for which some $C_{j_*,k}$ is nonzero. Multiply by $(z - 2^n)^{j_*}$ and evaluate the resulting polynomial identity at $z = 2^n$. This evaluates a cleared polynomial identity, not the analytic series at a branch point. All lower- j terms vanish, giving

$$\sum_{k=0}^K C_{j_*,k}(2^n) n^k = 0 \quad (n \in A).$$

The polynomial $R(X, Y) = \sum_{k=0}^K C_{j_*,k}(Y) X^k$ is nonzero. This contradicts Lemma 9.1, because A is infinite. The same proof covers $j_* = 0$: the polynomial identity itself may simply be evaluated. $\square$

9.3 The mixed-jet theorem

Theorem 9.3 (Hereditary mixed differential independence). *For every infinite $A \subseteq \mathbb{N}_{>0}$, the family*

$$\{\partial^j \delta^k F_A : j, k \geq 0\} \quad (9.2)$$

is algebraically independent over $B = \mathbb{C}(z)((t^{\mathbb{Q}}))$. Equivalently, the homomorphism from the free differential polynomial algebra in one indeterminate with two commuting derivations to H , sending that indeterminate to F_A , is injective.

Proof. Any finite collection of mixed jets is contained in a rectangle $0 \leq j \leq J, 0 \leq k \leq K$. By (8.6), the vector of all jets in the rectangle has the form of Theorem 5.1, with $r_n = h_n, \gamma_n = n$, and coefficient vectors from Lemma 9.2. Lemma 8.1 supplies square-class independence, and Lemma 9.2 supplies full cofinite span. Thus $E = \emptyset$ and the entire finite rectangle is algebraically independent over B . This proves the infinite family assertion. The formal mixed jets in a differential polynomial algebra with commuting derivations are indexed by pairs (j, k) ; their evaluation is injective exactly when the family (9.2) is algebraically independent. $\square$

This excludes every finite-order algebraic partial differential equation with coefficients in B , not just linear equations, autonomous equations, or equations of some bounded order. Setting $k = 0$ proves ordinary differential transcendence in z ; setting $j = 0$ proves differential transcendence for the scale derivation. Because $K_{\mathbb{Q}}(z) \subseteq B$, the result is stronger than independence over the rational function field with surcomplex coefficients from $K_{\mathbb{Q}}$.

Remark 9.4 (The upgrade a companion warning declines). The companion report on dynamics and normal forms [4] proves an algebraicity dichotomy for its Schröder-type solutions H_ε : for $V \in \mathbb{C}[w]$ with $V(0) = 0$ and $V'(0) \neq 0$, the solution H_ε is algebraic over $\mathcal{B} = \mathbb{C}(w)((t^\Gamma))$ if and only if $\deg V = 1$, so it is transcendental over $\mathcal{B}$ whenever $\deg V \geq 2$. Its warning `dyn:warn:diffalg` deliberately refuses to upgrade that to “differentially transcendental”, because its own H_ε satisfies the first-order linear equation $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$ with $\Omega_\varepsilon \in \mathcal{B}$. That refusal is correct and is not weakened by anything here: a function with a first-order linear equation over the base is differentially algebraic over it, whatever else is true.

Theorem 9.3 supplies the upgrade for *different* functions over a base of the same shape. With $k = 0$ it says that $\{\partial_z^j F_A : j \geq 0\}$ is algebraically independent over $\mathbb{C}(z)((t^{\mathbb{Q}}))$, that is, F_A satisfies no algebraic differential equation of any finite order over that field — the exact conclusion `dyn:warn:diffalg` declines for H_ε . The base is the $\Gamma = \mathbb{Q}$ case of the same $\mathbb{C}(z)((t^\Gamma))$ shape, and by Theorem 10.1 the statement holds for continuum many functions on the one fixed disk $D = \{|z| < 1\}$, with no shrinking of D as the parameter varies. The two reports therefore describe different functions, not a disagreement: the companion’s H_ε remains differentially algebraic over $\mathcal{B}$, and F_A is not. The contrast is sharpened rather than softened by (9.3) below, where an F_A that satisfies a clean dilation functional equation is nevertheless differentially transcendental.

As in the actual-surreal application, the same finite jet families remain independent after any algebraic base extension of B in the ambient field. This follows from invariance of transcendence degree under algebraic base extension, not from a claim that the original sign changes fix that enlarged base.

9.4 A functional equation does not supply a differential equation

Take $F = F_{\mathbb{N}_{>0}}$. Reindexing a Hahn-summable family gives the exact identity

$$F(2z, t) - tF(z, t) = t\sqrt{1-z}, \quad |z| < 1. \quad (9.3)$$

Indeed, the $n = 1$ term of $F(2z, t)$ is $t\sqrt{1-z}$, and its remaining terms are $tF(z, t)$. The expression $F(2z, t)$ is legitimate here because the coefficients of F are holomorphic on D_2 .

Thus a simple dilation relation and quadratic coefficient functions coexist with complete mixed-jet algebraic independence. Dilation is an additional operator; replacing it by a finite collection of derivatives would be an invalid inference. The result does not exclude functional equations involving shifted or dilated arguments.

9.5 Rational Taylor data and a summation warning

From (8.4), coefficientwise Taylor extraction yields

$$\partial^j F(0, t) = (-1)^j c_j \sum_{n \geq 1} 2^{-nj} t^n = (-1)^j c_j \frac{t}{2^j - t} \in \mathbb{C}(t). \quad (9.4)$$

Every specialized mixed jet $\partial^j \delta^k F(0, t)$ is therefore rational in t as well. In the ordinary formal power-series ring $K_{\mathbb{Q}}[[z]]$, the Taylor image of F is

$$\sum_{j \geq 0} (-1)^j \binom{1/2}{j} \frac{t}{2^j - t} z^j. \quad (9.5)$$

To verify this equality, fix j , extract the ordinary complex Taylor coefficient of each h_n , and then sum the Hahn family indexed by n . This is a coefficient-by-coefficient construction of a formal z -series.

Warning 9.5. For an ordinary nonzero complex value of z , expression (9.5) is *not* a Hahn sum indexed by j : infinitely many of its terms contribute to exponent t^1 . Its meaning is as a formal Taylor series in z , or as a jointly summable expression when z is a suitable infinitesimal. One must not interchange the two infinite summation notions without a support certificate.

There is also no contradiction between (9.4) and Theorem 9.3. The theorem is about functions in z over the specified function field. Evaluation at a point need not preserve algebraic independence. In fact, evaluation at any ordinary $z_0 \in D$ places these values inside $K_{\mathbb{Q}}$ itself. Unlike the actual numbers of Section 7, their point values are not being asserted transcendental over that field.

10 A continuum of jointly free surcomplex analytic jets

Use the almost disjoint sets A_{α} from (7.7) and define

$$G_{\alpha}(z, t) = F_{A_{\alpha}}(z, t) = \sum_{n \in A_{\alpha}} t^n \sqrt{1 - z/2^n}.$$

Every G_{α} has the same common holomorphic coefficient disk D . No shrinking of the domain is required as α varies.

Theorem 10.1 (Continuum many jointly independent mixed jets). *The family*

$$\{\partial^j \delta^k G_{\alpha} : \alpha \in \{0, 1\}^{\mathbb{N}}, j, k \geq 0\} \quad (10.1)$$

is algebraically independent over $B = \mathbb{C}(z)((t^{\mathbb{Q}}))$. It embeds the free differential polynomial algebra on c generators, with two commuting derivations, into $H = L((t^{\mathbb{Q}}))$. Moreover $\text{trdeg}_B H = c$; the constructed family has the largest possible cardinality for algebraic independence in this workspace.

Proof. Choose finitely many distinct $\alpha_1, \dots, \alpha_r$ and a finite rectangle of jet orders. At each coefficient index n , concatenate the jet coefficient vector of Lemma 9.2 in block a if $n \in A_{\alpha_a}$, and use a zero vector in that block otherwise. For each a , its infinite private index set contributes only to block a . After any finite deletion, Lemma 9.2 shows that these vectors span the entire

block. The cofinite span of the combined vectors is consequently the full combined jet space. Theorem 5.1 proves algebraic independence of the chosen finite tuple, and hence of all of (10.1).

A differential polynomial uses only finitely many generators and mixed jets, giving the claimed differential-algebra embedding. Finally, $|\mathbb{C}(z)| = \mathfrak{c}$, the algebraic extension L has cardinality $\mathfrak{c}$, and $\mathbb{Q}$ is countable. Thus $|H| \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c}$. The exhibited independent family has cardinality $\mathfrak{c}$, proving the exact transcendence degree and the maximality assertion. $\square$

Remark 10.2 (No assertion of a transcendence basis). The cardinal equality does not say that the displayed jet family is a transcendence *basis*, or that H is algebraic over the field it generates. It says that the family is independent and has maximal possible cardinality. Proper subsets of an infinite basis can have the same cardinality as the basis, so cardinal maximality must not be confused with maximality under inclusion.

11 From coefficient functions to surcomplex halos

11.1 Support-certified infinitesimal substitution

An actual surcomplex halo value may involve scales beyond $\mathbb{Q}$. Let Γ be a set-sized ordered subgroup of the surreal exponent group containing $\mathbb{Q}$, and put $K_\Gamma = \mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i]$ via $t^\gamma = \omega^{-\gamma}$. Let $z_0 \in D$ be an ordinary complex number and let $\varepsilon \in K_\Gamma$ be infinitesimal, meaning either $\varepsilon = 0$ or $\text{supp}(\varepsilon) \subset \Gamma_{>0}$. Define

$$F_A(z_0 + \varepsilon, t) = \sum_{n \in A} \sum_{\ell \geq 0} t^n \frac{h_n^{(\ell)}(z_0)}{\ell!} \varepsilon^\ell. \quad (11.1)$$

Proposition 11.1 (Joint summability of the halo formula). *Formula (11.1) is a Hahn-summable double family and gives an element of K_Γ . The same holds for every mixed jet and every finite polynomial combination of the functions considered above.*

Proof. The case $\varepsilon = 0$ reduces to the original sum. Otherwise write $T = \text{supp}(\varepsilon) \subset \Gamma_{>0}$ and $S = A \subset \mathbb{N}_{>0}$. By the positive-support Neumann lemma, the monoid T^* generated by T , including zero, is well ordered. For each exponent it has only finitely many representations as a word in T , counting all possible word lengths. Therefore the family $(\varepsilon^\ell)_{\ell \geq 0}$ is Hahn summable. The set $S + T^*$ is well ordered, with finitely many pairs contributing at any fixed exponent. These two finiteness statements together give joint summability of the (n, ℓ) family. The complex Taylor coefficients do not enlarge any support. Derivatives of h_n are holomorphic at the same z_0 , and finite products use the same support lemmas, proving the remaining assertions. $\square$

This is a direct certificate in an explicitly chosen set-sized Hahn workspace. It is not a convergence assertion for partial sums in the fine topology of the entire class $\mathbf{No}[i]$, and it makes no claim that F_A is defined at every infinite surcomplex argument.

11.2 Functional independence is visible on ordinary points

Although the largest coefficient field B is a formal meromorphic-germ Hahn field, the independence conclusion has an ordinary-point functional meaning over $K_{\mathbb{Q}}(z)$.

Proposition 11.2. *No nonzero polynomial over $K_{\mathbb{Q}}(z)$ in finitely many of the mixed jets of Theorem 10.1 vanishes identically on their common ordinary disk of definition. The same polynomial cannot vanish identically on all the corresponding halo values.*

Proof. Clear the finitely many rational-function denominators, obtaining a nonzero polynomial with coefficients in $K_{\mathbb{Q}}[z]$. Its evaluation on the chosen jets is a nonzero element of H by

Theorem 10.1. In fact it lies in $\mathcal{O}(D)((t^{\mathbb{Q}}))$: only finite products and sums of common-domain coefficient functions have been used. At least one of its Hahn coefficients is therefore a nonzero holomorphic function on D . It is nonzero at some ordinary point, and indeed outside its discrete zero set.

For each denominator in $K_{\mathbb{Q}}[z]$, the coefficient of its lowest valuation is a nonzero ordinary complex polynomial. Away from the finite union of their zeros, the denominator does not specialize to zero. Choose a point where the numerator coefficient is nonzero and these finitely many denominator coefficients are nonzero. The original polynomial expression is nonzero there. Ordinary points are halo points with $\varepsilon = 0$, proving the last assertion. $\square$

11.3 Three differentiations that must not be conflated

In Section 7, D_{BM} acts on actual surreal numbers and satisfies $D_{\text{BM}} = -t\delta$ on the stated rational-monomial fields. In the analytic coefficient field, ∂_z differentiates the independent complex parameter, while δ acts on the Hahn scale and fixes the coefficient functions. These are two explicitly defined commuting operators on a function field.

Specializing z to a nonconstant surreal expression can introduce a chain-rule term under D_{BM} . It is therefore incorrect to identify ∂_z with D_{BM} , or to assert without qualification that the function-field derivations remain those same operators after every surreal substitution. The algebraic independence theorem is formulated before such substitutions; Section 7 separately supplies a direct, normalized Berarducci–Mantova result for actual numbers.

12 Algebraic approximation depends on the value group

Theorem 3.1 also gives a sharp approximation criterion. It explains how a transcendental series can be approximated arbitrarily closely by algebraic ones in one workspace, but not after the value group is enlarged.

Theorem 12.1 (Closure of the algebraic coefficient part). *Let $B = M((t^{\Gamma}))$, let $\overline{M}$ be an algebraic closure in characteristic zero, and let $\mathcal{A}$ be the relative algebraic closure (3.2). A series $f = \sum_{\eta} c_{\eta} t^{\eta} \in \overline{M}((t^{\Gamma}))$ lies in the valuation-topological closure of $\mathcal{A}$ if and only if*

$$[M(c_{\eta} : \eta \leq \gamma) : M] < \infty \quad \text{for every } \gamma \in \Gamma. \quad (12.1)$$

Zero coefficients may be omitted from the generated field.

Proof. The valuation neighborhoods of f are the sets $\{g : v(f - g) > \gamma\}$ for $\gamma \in \Gamma$. If $g \in \mathcal{A}$ belongs to this neighborhood, its coefficients through exponent γ agree with those of f . All coefficients of g lie in one finite extension of M by Theorem 3.1, proving necessity.

Conversely, suppose (12.1). The truncation $f_{\leq \gamma} = \sum_{\eta \leq \gamma} c_{\eta} t^{\eta}$ has well-ordered support and coefficients in the finite extension from that condition. It lies in $\mathcal{A}$, and either equals f or its difference from f has least exponent strictly greater than γ . It approximates f in the required neighborhood, proving sufficiency. $\square$

For $\Gamma = \mathbb{Q}$, the series $\sum_{n \geq 1} \sqrt{p_n} t^n$ satisfies the criterion: below any rational cutoff there are only finitely many of its positive integer exponents. It is therefore a valuation limit of its finite algebraic truncations, despite its differential transcendence.

For a contrasting example, take $\Gamma = \mathbb{Z}^2$ in lexicographic order and

$$f = \sum_{n \geq 1} \sqrt{p_n} t^{(0,n)}.$$

Its support is still well ordered and has order type ω . However, every one of its exponents lies below $(1, 0)$. At that cutoff the coefficient field is the infinite multiquadratic extension, so the criterion fails. No algebraic element approximates f with error valuation greater than $(1, 0)$.

Under the embedding $(a, b) \mapsto a\omega + b$ of value groups, the second series has the same displayed surreal expansion $\sum_n \sqrt{p_n}\omega^{-n}$. The enlarged workspace has an additional precision scale $\omega^{-\omega}$ which no finite truncation reaches. Thus the topology and the available precision scales must be specified; Hahn summability alone is not valuation convergence of finite truncations in an arbitrary-rank field.

13 Effective limitations and exact verification

13.1 Algebraicity is undecidable for unrestricted computable streams

Consider the representation

$$x_\beta = \sum_{n \geq 1} \beta_n \sqrt{p_n} t^n, \quad \beta_n \in \{0, 1\},$$

where a program is promised to compute the total sequence (β_n) . Every coefficient has a finite exact description; the support is contained in the positive integers. Hence no support-validity problem is hidden in this representation.

Proposition 13.1. *There is no algorithm that, for every such total computable coefficient stream, decides whether x_β is algebraic over B_0 . There is also no algorithm deciding whether it is differentially algebraic over B_0 for D_{BM} .*

Proof. Given a program e on a fixed input, define β_n by simulating e for n steps: return 1 if it has not yet halted and 0 otherwise. This coefficient program is total regardless of whether e halts. If e halts, only finitely many β_n are nonzero, so x_β is algebraic and therefore differentially algebraic. If e never halts, $x_\beta = \xi_{\mathbb{N}_{>0}}$, which is differentially transcendental by Theorem 7.1, and in particular transcendental. Either proposed decision procedure would decide whether arbitrary programs halt. Such a procedure is impossible by the standard diagonal argument: a program that halts exactly when the purported halting test says it does not halt contradicts the test applied to its own code. $\square$

Every finite set of coefficient observations is also compatible with both a finite-support algebraic continuation and an infinite-support differentially transcendental continuation. The latter assertion uses the fact that *every* infinite subset works, not just the full set of positive integers. Thus the finite certificates of Corollary 6.1 require structural information about the stream; numerical or symbolic inspection of a long prefix is not a substitute for such information.

13.2 What the accompanying checks do

The supplied `verify.py` uses exact arithmetic and symbolic identities. It checks the top-degree finite-difference formula, small multiquadratic sign products, the quadratic exceptional relation, Vandermonde and mixed-jet ranks, the derivative formulas, the finite boundary term in the dilation identity, Taylor coefficient identities, and the finite behavior of the binary-prefix construction. All 404 recorded checks passed with Python 3.13.5 and SymPy 1.14.0; the exact environment is also recorded in `verification_results.json`.

These checks are deliberately subordinate to the proofs. A finite rank calculation does not prove the hereditary full-span lemma for all orders and all infinite subsets; that lemma is proved by the pole argument and Lemma 9.1. Finite sign-orbit examples do not prove the

classification for arbitrary well-ordered supports; that theorem is proved by exact Galois actions, finite differences, and cofinite spanning. No Lean file is included, and no machine-checked theorem is claimed.

13.3 Proof dependencies and falsifiable checkpoints

The core proof has a short dependency structure. Finite multiquadratic Galois theory gives independently available sign changes. A degree- D polynomial relation then forces its highest polarization to vanish on every distinct-index D -tuple. Cofinite spanning forces that polarization to vanish identically. Finally, one transverse linear functional recovers exactly the finite exceptional multiquadratic extension. Each of these steps is isolated in a separately stated lemma or theorem.

The two applications add different spanning arguments. Actual surreal Euler jets use the Vandermonde determinant. Analytic mixed jets use the largest pole order and the impossibility of a nonzero polynomial vanishing on infinitely many points $(n, 2^n)$. The continuum constructions add only the finite-overlap property of binary-prefix sets.

The assumptions have specific roles. Distinct support exponents ensure that sign-conjugate primitive elements are distinguishable coefficientwise. Independent square classes license independent sign changes. Characteristic zero ensures that the top polarization is detected by $D!$ and that separability causes no coefficient-orbit pathology. Finite-dimensional vector spaces make the cofinite span stabilize after finitely many deletions. Dropping any of these hypotheses requires a new argument, not a notational adjustment.

14 Further questions and final assessment

The article establishes an exact relation theory for one substantial class of Hahn constructions. It does not classify arbitrary algebraic coefficient families. When the available coefficient Galois actions are correlated rather than independent sign changes, their orbits need not supply the independent translation directions used in the proof. A natural next problem is to replace the square-root hypotheses with a usable condition on finite Galois representations, and to determine which parts of the affine-plane classification survive.

A second problem is effectivity under certificates. For a stream supplied with a finite-state recurrence, a certified pole pattern, or an explicit cofinite-spanning proof, one can ask for algorithms that compute its transcendence dimension and exceptional field. The unrestricted decision problem of Proposition 13.1 does not rule out such structured subclasses. Corollary 6.1 suggests a bounded-degree certificate format based on disjoint spanning sets and a nonzero mixed difference.

A third problem concerns the interaction with richer surreal derivations and larger monomial groups. The actual-number theorem here deliberately uses rational exponents, where the normalized Berarducci–Mantova restriction is explicit. Extending its transparent jet-spanning proof to supports involving general surreal exponents would require control of their derivatives, not just their order. Likewise, the analytic theorem does not automatically survive substitutions in which the parameter itself carries a nontrivial surreal derivative.

The established conclusions are already stronger than a collection of isolated transcendence examples. The cofinite span determines the whole generated finite-type algebra, the finite exceptional field, and the geometric splitting of every relation. The same mechanism constructs a continuum of positive infinitesimal surreal numbers with all derivatives independent over a full rational-coefficient Hahn field, and a continuum of common-domain surcomplex analytic functions with all mixed jets independent. These are precise, nontrivial research results offered with complete proofs and explicit limitations, while their bibliographic novelty remains a matter for independent review.

A Notation and scope at a glance

Symbol	Meaning and scope
$M((t^\Gamma))$	Hahn field: coefficients in M , well-ordered support in the set-sized ordered abelian group Γ .
W	Cofinite span of coefficient vectors; intersection over finite deletions, not arbitrary ordinal tails.
E	Finite set of vectors outside W ; precisely the exceptional coefficient indices in the classification.
B_E	$M(r_i : i \in E)((t^\Gamma))$; the exact algebraic part of $B(Y)$, of degree $2^{ E }$ over B .
$\mathbb{A}_{\mathbb{R}}$	Real-algebraic numbers, a countable subfield of $\mathbb{R}$.
$B_0 \subset H_0$	$\mathbb{Q}((t^{\mathbb{Q}})) \subset \mathbb{A}_{\mathbb{R}}((t^{\mathbb{Q}}))$, concretely embedded in $\mathbf{No}$ by $t = \omega^{-1}$.
D_{BM}	Normalized Berarducci–Mantova derivation on actual surreals; its restriction to H_0 is $-t\delta$.
δ	Euler derivation; $\delta(t^q) = qt^q$ and coefficient-field elements are fixed.
∂	In analytic sections, ordinary differentiation in the independent parameter z on coefficient germs.
B in analytic sections	$\mathbb{C}(z)((t^{\mathbb{Q}}))$, a formal relation field, not a common-domain holomorphic ring.
$K_{\mathbb{Q}}$	$\mathbb{C}((t^{\mathbb{Q}})) \subset \mathbf{No}[i]$; $K_{\mathbb{Q}}(z)$ is a smaller relation field contained in B .
D	The ordinary complex disk $ z < 1$, shared by every analytic coefficient in every constructed function.
c	The cardinal $2^{\aleph_0}$, not the proper-class size of all surreals.

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- [3] Vladimir Reshetnikov, *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers*, companion report in this collection, docs/surcomplex/differential-equations/. Its section “The rational-exponent workspace” fixes $\mathbb{C}((t^{\mathbb{Q}}))$ with $t = \omega^{-1}$ at `diff:eq:HahnQ`, and its Proposition `diff:prop:hahnderiv` proves the derivative formula `diff:eq:hahnderiv` on a real-exponent workspace. That report is the statement of record for those two facts; they are cited, not reproved, in Section 7. Its warning `diff:warn:monomialmap` separates the Conway monomial map from general surreal exponentiation.
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